AI Anxiety Map: Geographic and Emotion Analysis of Global Labor-Market News, 2021–2024

Amit Ner Gaon
Department of Software and Information Systems Engineering
Ben-Gurion University of the Negev
Beer-Sheva, Israel
amitner@post.bgu.ac.il

***Abstract*—Generative artificial intelligence (AI) has intensified debate about job displacement and new economic opportunities, but global evidence about how this debate is framed remains limited. This study analyzes labor-market news in the GDELT Global Knowledge Graph [1] from 2021 through 2024. It compares two equal periods before and after AI breakthrough at January 1, 2023 using anxiety and opportunity scores constructed from global content sentiment analysis, geographic summaries, and location-emotion similarity networks. Aggregated location-theme article occurrences increased by 13.9% after the AI cutoff, while mean anxiety and opportunity scores each decreased by about 6.9%. This global reduction was not geographically uniform: anxiety increased in 122 of 197 countries and in 46 of 50 U.S. states. Country changes in anxiety and opportunity were positively correlated ($r = 0.57$), indicating that they often represent common variation in emotional intensity rather than opposite ends of one scale. Among 18 high-volume locations shared by both network periods, mean pairwise emotional similarity decreased by 1.5%, with 80 pairs converging and 73 diverging. The findings demonstrate the value of combining temporal, geographic, and network views, for large scale emotion analysis.**

Code is available at:
`github.com/AmitNG2000/Data-Mining-Assignments`.

***Index Terms*—generative AI, labor market, GDELT, emotion analysis, geographic analysis, network analysis, News**

I. Introduction

Generative AI can automate parts of knowledge work while also complementing workers and creating new tasks. Research on the history of automation describes the same broader tension: technology replaces some tasks, increases the value of complementary skills, and distributes adjustment costs unevenly [2], [3]. Recent research reports productivity gains from generative AI across a wide range of workplace tasks [4], [5]. These productivity gains often raise concerns about job displacement, downward pressure on wages, and the obsolescence of existing skills.

News media provide a large, time-stamped record that can serve as an proxy for the concerns, priorities, and emotional responses expressed within communities [6]. Although news coverage does not directly measure public opinion or workers' experiences, it enables large-scale and systematic collection of discourse data across countries, as well as comparative sentiment and emotion analysis. Prior research has shown that news coverage of AI combines persistent optimism with growing concern about ethics, control, and employment [7]. This research extends this approach to a multinational news and models geographic variation and cross-location similarity.

January 1, 2023 is used as an operational marker for the rapid diffusion of generative AI: ChatGPT was released in late 2022, GPT-4 followed in March 2023, and generative-AI tools achieved substantial organizational adoption during 2023 [8]–[10].

This study addresses three research questions:

1) **RQ1:** How did anxiety, opportunity, and overall sentiment framing of the global labor-market differ between the BEFORE AI and AFTER AI periods?
2) **RQ2:** How did the emotional framing of the labor-market vary across countries between the BEFORE AI and AFTER AI periods?
3) **RQ3:** How similar or polarized were locations in their emotional framing of labor-market, and how did these relationships change between the BEFORE AI and AFTER AI periods?

This study presents a reproducible framework combining emotion analysis, temporal comparisons, geographic summaries, and similarity networks.

II. Related Work

Recent studies show that generative AI can increase productivity by improving both the speed and quality of work. These benefits vary across tasks and workers, with particularly strong gains among less-experienced employees [4], [5].

At the same time, public attitudes toward AI include substantial concern about employment risk. Survey evidence from 16 Latin American countries shows that fear of job loss due to AI varies over time and across countries [11].

Although it does not directly measure public opinion, content analysis of digital discourse can reveal how AI is emotionally framed on the public. Research on Reddit discussions identifies both optimism and "algorithmic anxiety" related to job displacement, professional identity, and trust in employers [12]. Similarly, an analysis of `The New York Times` found that AI coverage was generally more optimistic than pessimistic, while concerns about employment, ethics, and loss of control increased over time [7].

III. Methodology

A. Collection and Aggregation

The pipeline processes both the original-language and translated streams of the GDELT 2.1 Global Knowledge Graph (GKG). A record is retained when its structured `V2Themes` field contains at least one predefined labor-market theme, such as employment, unemployment, jobs, workers, hiring, layoffs, wages, income, careers, unions, gig work, or freelance work. For each retained record, the pipeline extracts the publication date, matched themes, location, tone, and emotions.

Because a single GKG record may mention multiple locations and match multiple labor-market themes, the pipeline creates one observation for each location–theme combination. These observations are then aggregated into seven-day intervals by location, country code, administrative code, and theme. Accordingly, `article_count` represents the number of grouped record occurrences rather than the number of globally unique articles.

B. Emotional Scores and Period Comparison

GDELT's Global Content Analysis Measures (GCAM) apply multiple dictionary-based content-analysis systems to news articles, including LIWC and WordNet-Affect, to identify emotional and cognitive language. LIWC classifies words into psychologically meaningful categories, while WordNet-Affect organizes words and concepts associated with emotions such as anxiety and fear [13]–[15].

For aggregate row i, the anxiety score is calculated as the mean of the available LIWC anxiety, WordNet-Affect anxiety, WordNet-Affect fear, and GCAM c6.6 uncertainty dimensions:

$$A_i = \frac{1}{|\mathcal{N}_i|} \sum_{d \in \mathcal{N}_i} x_{i,d}. \tag{1}$$

The opportunity score averages the available WordNet hope and optimism dimensions:

$$O_i = \frac{1}{|\mathcal{Q}_i|} \sum_{d \in \mathcal{Q}_i} x_{i,d}. \tag{2}$$

The anxiety-opportunity gap is $F_i = A_i - O_i$.

Rows whose window starts before January 1, 2023 are assigned to the before-AI period, and all later rows to the after-AI period. Global differences are assessed with Welch tests, while weekly series are used to inspect whether a period average conceals trends or peaks. Country and U.S. state summaries are computed only for units observed with both scores in both periods. Pearson correlation between country-level changes in A and O are evaluated.

C. Location-Emotion Networks

For each period, a weighted bipartite graph links up to 20 countries with the largest interaction volumes to nine GCAM emotion dimensions. An edge weight is the mean score for that location-emotion combination. The projected similarity between locations u and v is

$$S(u,v) = \sum_{e \in E(u) \cap E(v)} \min\{w(u,e), w(v,e)\}, \tag{3}$$

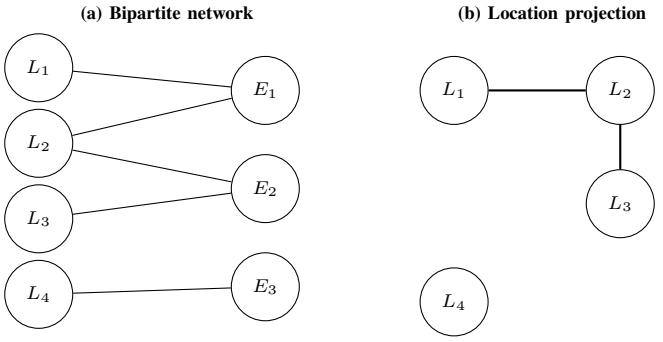


Fig. 1. Bipartite location–emotion network and its one-mode location projection. Two locations are connected in the projection if they share at least one emotion in the bipartite graph.

where $w(u,e)$ is the bipartite edge weight. This overlap measure is larger when two locations share emotions at similar intensities.

Figure 1 illustrates the projection of the location–emotion bipartite network onto a location-only network. Two locations are connected when both are associated with at least one common emotion.

D. Temporal Analysis

Temporal variation is examined at the same seven-day resolution used during data aggregation. For each location and interval, article occurrences and interactions are summed across labor-market themes, while anxiety and opportunity are calculated as the arithmetic means of the available location–theme scores. The global time series is then constructed by summing article occurrences and averaging the location-level scores within each interval.

Weekly anxiety and opportunity series are plotted with January 1, 2023 marked as the boundary between the two study periods. The series are inspected for trends, short-lived peaks, and whether any change near the cutoff is sustained. For geographic comparisons, each country's and U.S. state's mean score is calculated separately for the before- and after-AI periods, and temporal change is defined as the after-period mean minus the before-period mean.

IV. Experiments and Results

A. system Configuration

The production run covers January 1, 2021 through December 31, 2024, split into equal two-year periods. The analysis is implemented in Python 3.11 with pandas, SciPy, NetworkX, Matplotlib, Plotly, and PyDeck. Large-scale collection and notebook execution were run on the Ben-Gurion University Slurm cluster using CPUs.

B. Data

This study uses the GDELT 2.1 Global Knowledge Graph (GKG), a continuously updated international news database containing structured information on themes, locations, tone, and events [1], [15]. Its multilingual coverage, translated news

TABLE I
GLOBAL BEFORE-AFTER SUMMARY

Period	Occurrences	A	O	F
Before AI	718.9M	6.341	2.785	4.617
After AI	818.8M	5.902	2.592	4.225
Change (%)	+13.9	-6.9	-6.9	-8.5

TABLE II
DIRECTION OF COUNTRY-LEVEL SCORE CHANGES

Countries analyzed	Anxiety increased	Anxiety decreased	Opportunity increased	Opportunity decreased	Same direction	Opposite directions
197	122	75	78	119	137	60

stream, and GCAM emotion variables enable consistent cross-country analysis of labor-market discourse at a large scale. For this research, the data spans January 2021 to December 2024, providing two equal periods before and after the January 1, 2023 cutoff.

C. RQ1: Global Emotional Framing

Table I shows that global labor-market framing became less emotionally intense after the cutoff. Mean anxiety decreased from 6.341 to 5.902 (6.9%), mean opportunity decreased from 2.785 to 2.592 (6.9%), and the anxiety-opportunity gap decreased from 4.617 to 4.225 (8.5%). At the same time, the summed article counts across all location–theme groups increased by 99.9 million (13.9%). Because one article may contribute to multiple groups, this increase reflects greater indexed coverage rather than 99.9 million additional unique articles.

Welch tests reject equal period means for anxiety ($t = -88.44$), opportunity ($t = -124.40$), and the gap ($t = -64.04$), all $p < .001$. Figure 2 shows substantial variation within both periods and no clear discontinuity at January 1, 2023. The largest anxiety peak occurred before the cutoff, in July 2022 (12.017), whereas the largest after-period peak occurred in May 2023 (7.871).

D. RQ2: Cross-Country Emotional Variation

The global decline was not geographically uniform. As shown in Fig. 3 and Table II, anxiety increased in 122 of 197 countries and decreased in 75, whereas opportunity increased in 78 countries and decreased in 119.

Changes in anxiety and opportunity were positively correlated across countries ($r = 0.572$). Both scores increased in 70 countries and decreased in 67, giving 137 countries in which they moved in the same direction, compared with 60 in which they moved in opposite directions. At the marginal level, anxiety increased in most countries while opportunity decreased in most countries, but the frequent co-movement of the two scores shows that anxious and opportunity-oriented framing are not strict opposites.

E. RQ3: Location Similarity and Polarization

The period networks each contain 20 locations and all 190 possible pairwise edges; 18 locations occur in both periods, producing 153 directly comparable pairs. Among these shared pairs, mean similarity declined from 49.326 to 48.581, a decrease of 1.5%. The aggregate shift was therefore toward slightly less similar emotional framing. It was not uniform: 80 pairs became more similar and 73 became less similar. Figure 4 provides the complete weighted-network view for each period, while Fig. 5 isolates the largest matched-pair changes.

Figure 5 shows that the largest convergence was Russia–Ukraine (+97.7%), followed by Israel–Russia (+74.9%) and Israel–Ukraine (+72.7%). The largest divergence was China–Germany (-30.9%), followed by China–Vietnam (-26.8%) and Germany–Vietnam (-26.8%).

V. CONCLUSIONS

A. Discussion

The findings indicate that the emotional framing of global labor-market news became less intense after the January 1, 2023 cutoff. Mean anxiety and opportunity scores each declined by approximately 6.9%, while the anxiety-opportunity gap narrowed by 8.5%. However, the weekly time series remained highly variable and showed no abrupt discontinuity at the cutoff. These global trends also concealed substantial geographic variation: anxiety increased in 122 of 197 countries, while opportunity increased in 78, and changes in the two measures were positively correlated. At the network level, average emotional similarity among comparable location pairs decreased by only 1.5%. Since 80 pairs converged and 73 diverged, the results suggest a selective reconfiguration of emotional relationships across locations rather than widespread polarization.

A comparable survey covering a similarly broad set of countries throughout the entire study period is not available and would be difficult to construct retrospectively. Large-scale news data therefore provide a useful, time-stamped source for systematically comparing emotional framing across locations and periods using a consistent procedure.

B. Limitations

Several limitations should be acknowledged. First, the analysis treats the labor market as a broad domain and does not distinguish among industries, occupations, skill levels, or employment arrangements. As a result, the aggregate findings may obscure meaningful variation across labor-market segments. Second, the study does not provide detailed economic or political analyses of individual countries. Factors such as unemployment, inflation, regulation, elections, armed conflict, and differences in media systems may have contributed to the observed geographic and temporal patterns. Third, the analysis does not establish a causal relationship between the diffusion of generative AI and changes in emotional framing or cross-location similarity.

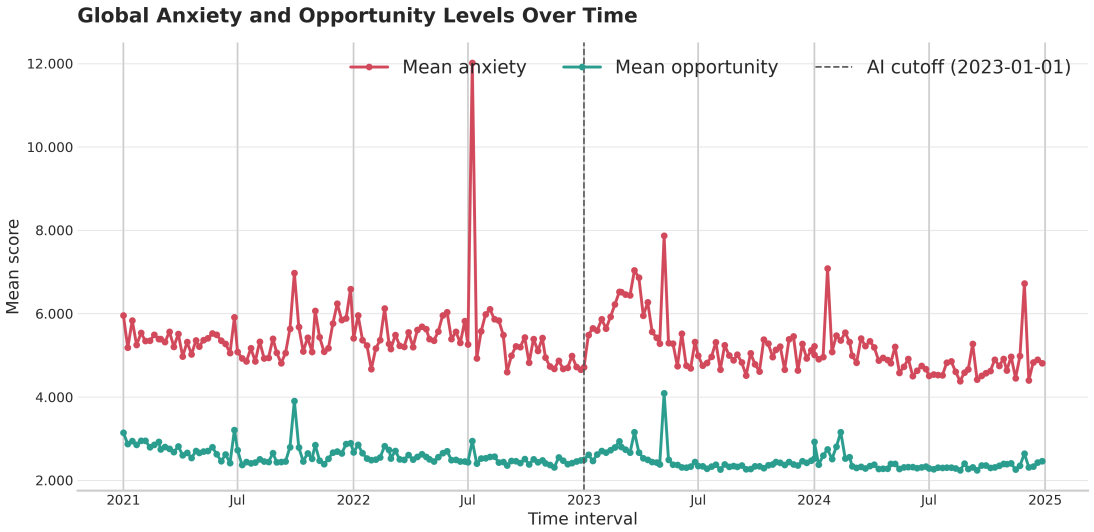


Fig. 2. Weekly global anxiety and opportunity scores. The dashed line marks the operational cutoff; the pre-cutoff peak cautions against a causal reading of the two-period comparison.

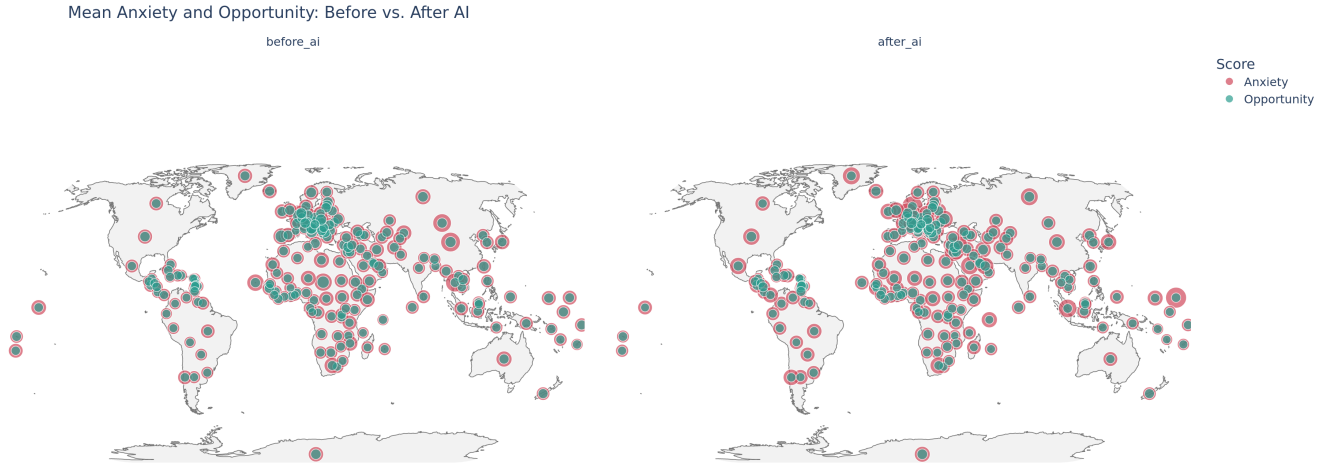


Fig. 3. Country-level mean anxiety and opportunity scores before and after the cutoff. Marker size represents the mean score; red denotes anxiety and green denotes opportunity.

C. Future Work

Future research should disaggregate labor-market discourse by industry, occupation, skill level, and employment arrangement to identify variation across labor-market segments. It could also incorporate economic, political, and survey data. In addition, causal inference methods could be applied to better identify the factors driving the observed changes.

Acknowledgment

The author acknowledges GDELT, the course staff, and Ben-Gurion University computing resources. OpenAI ChatGPT and Codex assisted with brain storming, coding and linguistic editing. The author verified the results and responsible for the analysis and conclusions.

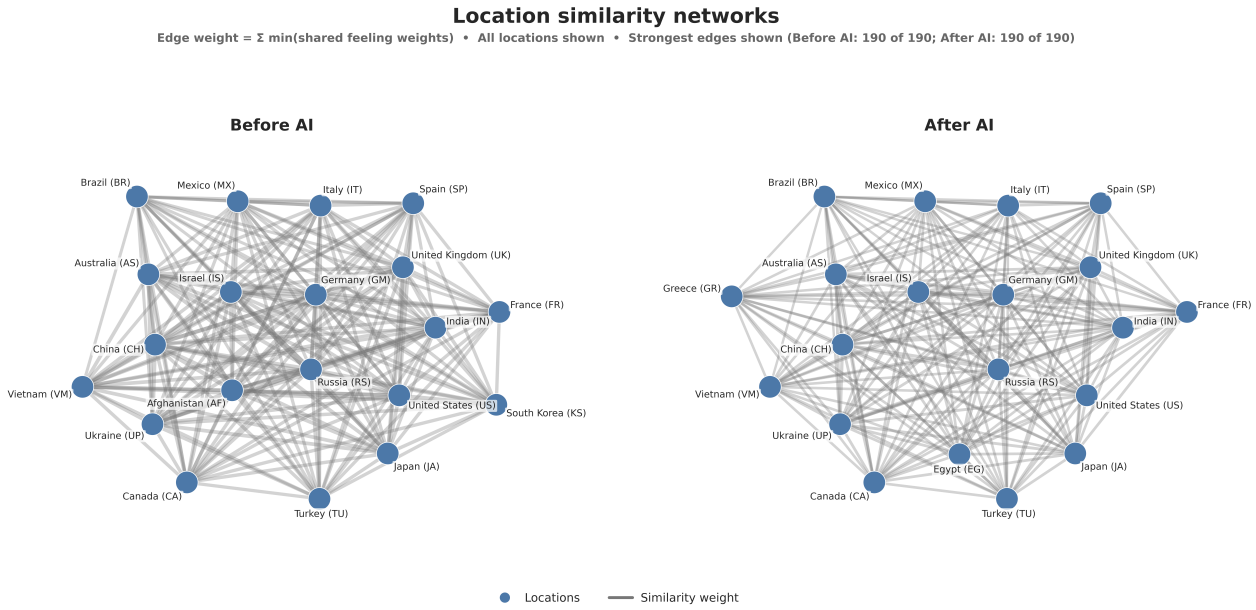


Fig. 4. Location-similarity networks before and after the cutoff. Nodes represent the 20 highest-volume locations in each period, and edge width represents the emotional-overlap similarity score. The layout supports visual comparison only; interpretation is based on the edge weights.

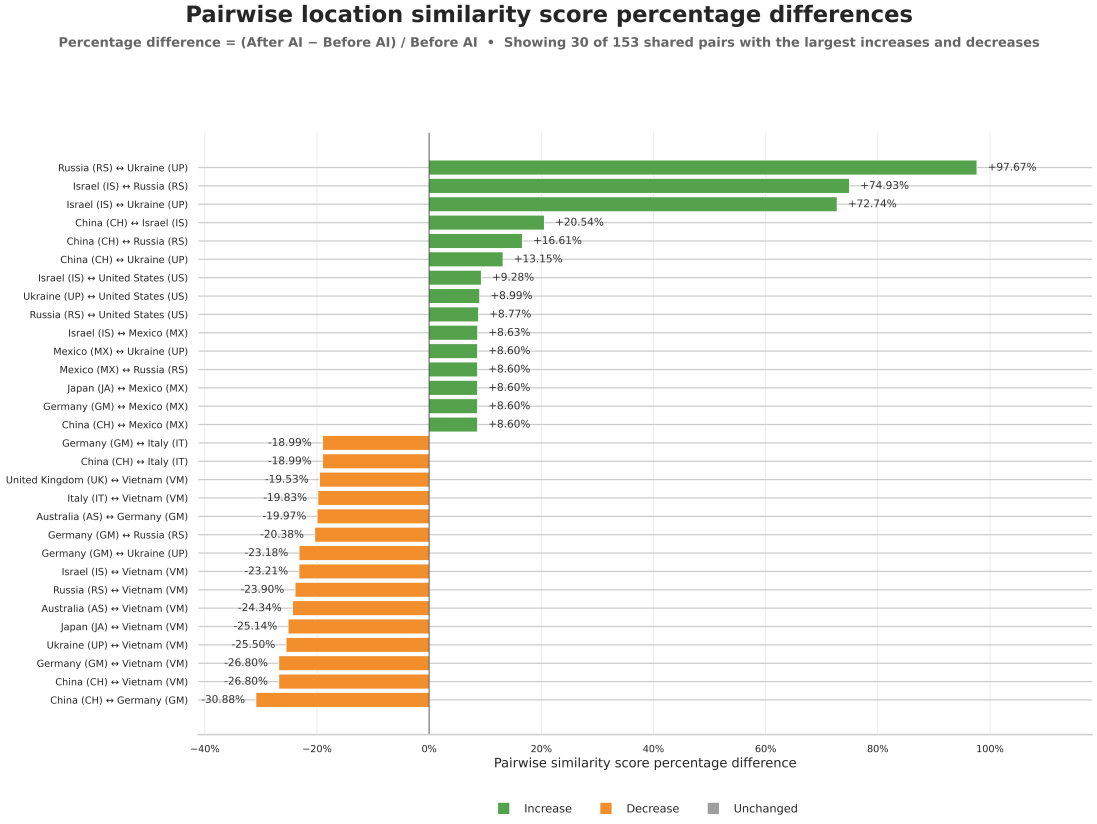


Fig. 5. Largest increases and decreases in pairwise emotional-similarity scores among the 153 location pairs observed in both periods. Positive values indicate convergence and negative values indicate divergence.

REFERENCES

[1] K. Leetaru and P. A. Schrodt, "Gdelt: Global data on events, location, and tone. isa annual convention," 2013.

[2] D. H. Autor, "Why are there still so many jobs? the history and future of workplace automation," *Journal of economic perspectives*, vol. 29, no. 3, pp. 3–30, 2015.

[3] D. Acemoglu and P. Restrepo, "Robots and jobs: Evidence from us labor markets," *Journal of political economy*, vol. 128, no. 6, pp. 2188–2244, 2020.

[4] S. Noy and W. Zhang, "Experimental evidence on the productivity effects of generative artificial intelligence," *Science*, vol. 381, no. 6654, pp. 187–192, 2023.

[5] E. Brynjolfsson, D. Li, and L. Raymond, "Generative ai at work," *The Quarterly Journal of Economics*, vol. 140, no. 2, pp. 889–942, 2025.

[6] G. Ploger, S. Soroka, C. Wlezien, and M. Traugott, "Time-varying relationships between the "sentiment" of us public opinion and news media," *International Journal Of Public Opinion Research*, vol. 37, no. 4, p. edaf052, 2025.

[7] E. Fast and E. Horvitz, "Long-term trends in the public perception of artificial intelligence," in *Proceedings of the AAAI conference on artificial intelligence*, vol. 31, no. 1, 2017.

[8] OpenAI, "Introducing chatgpt," https://openai.com/index/chatgpt/, Nov. 2022, accessed: 2026-08-01.

[9] ——, "Gpt-4," https://openai.com/index/gpt-4-research/, Mar. 2023, accessed: 2026-08-01.

[10] McKinsey & Company, "The state of ai in 2023: Generative ai's breakout year," https://www.mckinsey.com/capabilities/quantumblack/our-insights/the-state-of-ai-in-2023-generative-ais-breakout-year, Aug. 2023, accessed: 2026-07-25.

[11] A. Cremaschi, D.-J. Lee, and M. Leonelli, "Will ai take my job? evolving perceptions of automation and labor risk in latin america," in *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*, vol. 8, no. 1, 2025, pp. 653–664.

[12] A. Shekhar and M. D. Saurombe, "Algorithmic anxiety: Ai, work, and the evolving psychological contract in digital discourse," *Frontiers in psychology*, vol. 17, p. 1745164, 2026.

[13] Y. R. Tausczik and J. W. Pennebaker, "The psychological meaning of words: Liwc and computerized text analysis methods," *Journal of language and social psychology*, vol. 29, no. 1, pp. 24–54, 2010.

[14] C. Strapparava, A. Valitutti *et al.*, "Wordnet affect: an affective extension of wordnet." in *Lrec*, vol. 4, no. 1083-1086. Lisbon, Portugal, 2004, p. 40.

[15] The GDELT Project, "The GDELT global knowledge graph (GKG), version 2.1," Online, 2015. [Online]. Available: https://data.gdeltproject.org/documentation/GDELT-Global_Knowledge_Graph_Codebook-V2.1.pdf